Examiner only

3. *(a)* The first law of thermodynamics is given by:

$$\Delta U = Q - W$$

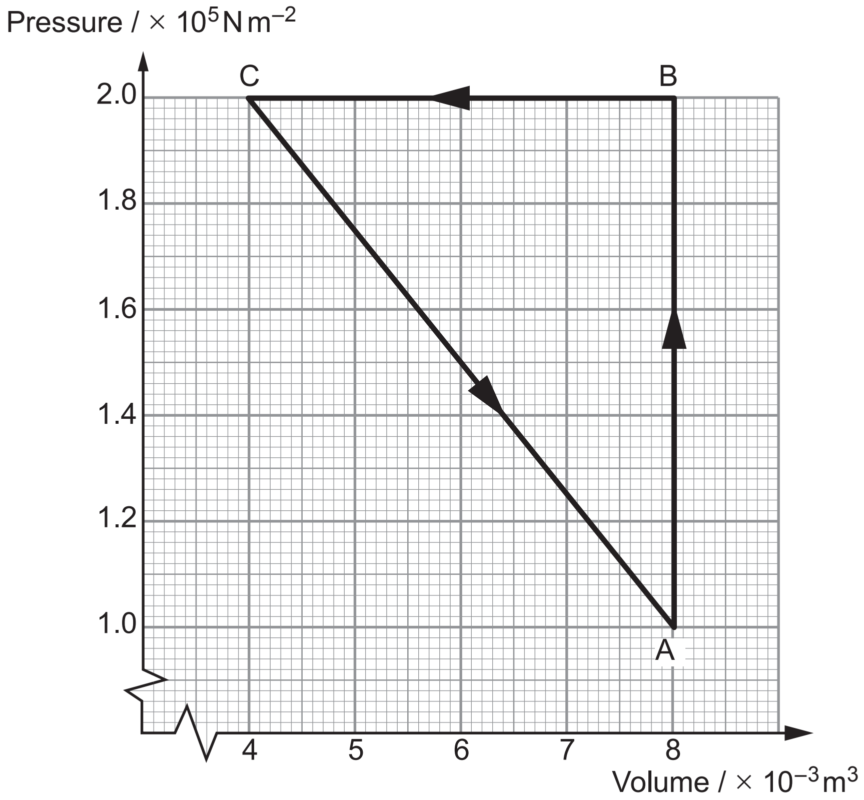
State what is represented by: [3]

ΔU ....

Q ....

W ....

(b) A fixed mass of gas is taken around the closed cycle A → B → C → A.



(i) Complete the table by describing each process in terms of pressure and volume. The description for process A → B is already inserted. For each process state if any work is done, and if so indicate if it is done *on* or *by* the gas. *No calculations are required.* [5]

Process	Description of process	Work done on / by gas (if any)
A → B	Increase in pressure at constant volume	
B → C		
C → A		

(ii) Determine the net work done on the gas during the entire cycle. [2]

Examiner only

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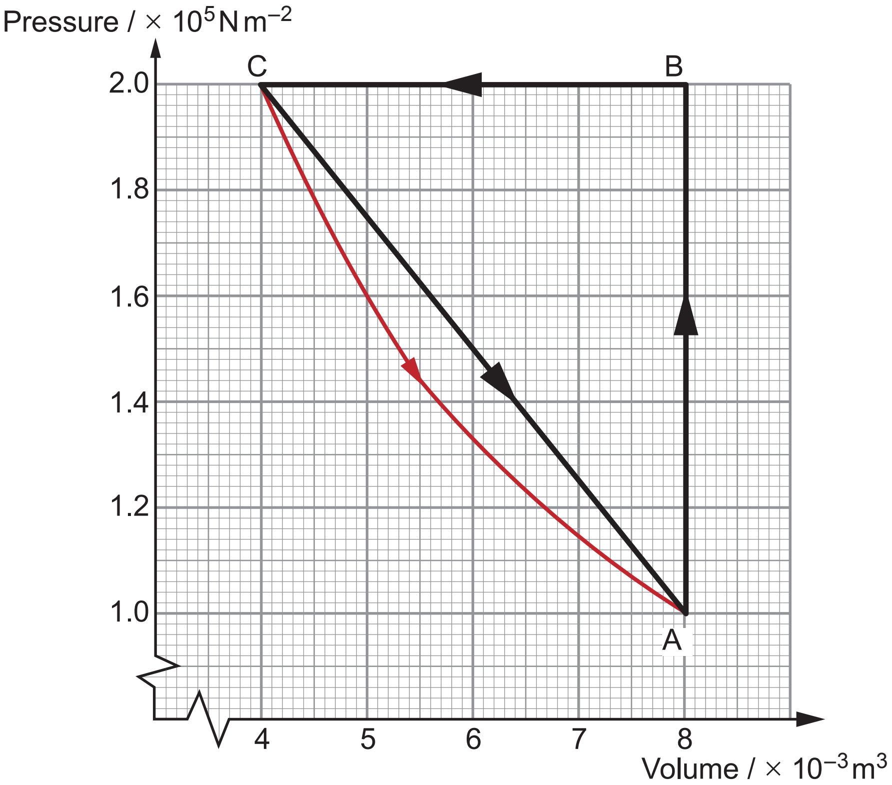
(c) The same mass of gas is taken around the cycle A → B → C → A for a second time. For this cycle the temperature is kept constant between C and A. The new path from C to A is shown in red on the graph below, together with the original path.



(i) Show that the temperature is constant along the new path from C to A. [3]

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(ii) Estimate the difference in the net work done in the two cycles. [2]

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